

## Solving simultaneous equations with graphs

Simultaneous equations can be solved by rearranging each equation so that it is expressed in terms of  $y$ , using a table to find sets of  $x$  and  $y$  coordinates for each equation, then plotting the graphs. The solution is the coordinates of the point where the graphs intersect.

### ▷ A pair of equations

This pair of simultaneous equations can be solved using a graph. Each equation will be represented by a line on the graph.

$$\begin{array}{r} 2x + y = 7 \\ -3x + 3y = 9 \end{array}$$

there are  $y$  terms in both equations

there are  $x$  terms in both equations

**To isolate  $y$  in the first equation,** rearrange the equation so that  $y$  is left on its own on one side of the equals sign. Here, this is done by subtracting  $2x$  from both sides of the equation.

$2x + y = 7$  is the first equation

$$\begin{array}{r} 2x + y = 7 \\ -2x \quad -2x \\ \hline y = 7 - 2x \end{array}$$

$-2x$  has also been added to this side

$-2x$  has been added to this side to cancel out the  $2x$  and isolate  $y$

**Find the corresponding  $x$  and  $y$  values** for the rearranged first equation using a table. Choose a set of  $x$  values that are close to zero, then work out the  $y$  values using a table.

the 7 does not depend on  $x$

choose a set of values for  $x$  that are close to 0

| $x$             | 1  | 2  | 3  | 4  |
|-----------------|----|----|----|----|
| 7               | 7  | 7  | 7  | 7  |
| $-2x$           | -2 | -4 | -6 | -8 |
| $y$<br>(7 - 2x) | 5  | 3  | 1  | -1 |

work out the value of  $-2x$  for each value of  $x$

the  $y$  value is the sum of 7 and  $-2x$

$7 - 6 = 1$

**To isolate  $y$  in the second equation,** rearrange so that  $y$  is left on its own on one side of the equals sign. Here, this is done by first adding  $3x$  to both sides, then dividing both sides by 3.

$-3x + 3y = 9$  is the second equation

$$\begin{array}{r} -3x + 3y = 9 \\ +3x \quad +3x \\ \hline 3y = 9 + 3x \\ \div 3 \quad \div 3 \quad \div 3 \\ \hline y = 3 + x \end{array}$$

$3x$  has been added to this side to cancel out  $-3x$ ; this isolates  $3y$

$3x$  must also be added to this side

divide both sides of equation by 3 to isolate  $y$

$3y \div 3 = y$

$9 \div 3 = 3$

$3x \div 3 = x$

**Find the corresponding  $x$  and  $y$  values** for the rearranged second equation using a table. Choose the same set of  $x$  values as for the other table, then use the table to work out the  $y$  values.

choose the same values of  $x$  as in the other table

| $x$            | 1 | 2 | 3 | 4 |
|----------------|---|---|---|---|
| 3              | 3 | 3 | 3 | 3 |
| $+x$           | 1 | 2 | 3 | 4 |
| $y$<br>(3 + x) | 4 | 5 | 6 | 7 |

the 3 does not depend on the value of  $x$

the value of  $+x$  is the same as  $x$

the  $y$  value is the sum of 3 and  $x$

$3 + 3 = 6$